

Iteration Method

To describe this method for finding a root of the equation we rewrite this equation in the form

$$x = \phi(x)$$

Now, let x_0 be an approximate root. Then, substituting in above Eq we get the first approximation as

$$x_1 = \phi(x_0)$$

Successive substitutions give the approximations

$$x_2 = \phi(x_1), \quad x_3 = \phi(x_2), \dots, \quad x_n = \phi(x_{n-1}).$$

The preceding sequence may not converge to a definite number. But if the sequence converges to a definite number ξ , then ξ will be a root of the equation $x = \phi(x)$. To show this, let

$$x_{n+1} = \phi(x_n)$$

be the relation between the n th and $(n+1)$ th approximations. As n increases, $x_{n+1} \rightarrow \xi$ and if $\phi(x)$ is a continuous function, then $\phi(x_n) \rightarrow \phi(\xi)$. Hence, in the limit, we obtain which shows that ξ is a root of the equation $x = \phi(x)$.

To establish the condition of convergence, we proceed in the following way we have

$$x_1 = \phi(x_0)$$

Then we get

$$\xi - x_1 = \phi(\xi) - \phi(x_0) = (\xi - x_0) \phi'(\xi_0), \quad x_0 < \xi_0 < \xi$$

Similarly, we obtain

$$\begin{aligned} \xi - x_2 &= (\xi - x_1) \phi'(\xi_1), & x_1 < \xi_1 < \xi \\ \xi - x_3 &= (\xi - x_2) \phi'(\xi_2), & x_2 < \xi_2 < \xi \\ &\vdots & \vdots \\ \xi - x_{n+1} &= (\xi - x_n) \phi'(\xi_n), & x_n < \xi_n < \xi \end{aligned}$$

If we assume $|\phi'(\xi_i)| \leq k$ (for all i) then

$$\left. \begin{aligned} |\xi - x_1| &\leq k |\xi - x_0| \\ |\xi - x_2| &\leq k |\xi - x_1| \\ |\xi - x_3| &\leq k |\xi - x_2| \\ &\vdots \\ |\xi - x_{n+1}| &\leq k |\xi - x_n| \end{aligned} \right\}$$

Multiplying the corresponding sides of the above equations, we obtain

$$|\xi - x_{n+1}| \leq k^{n+1} |\xi - x_0|$$

If $k < 1$, i.e., if $|\phi'(\xi_i)| < 1$, then the right side of above equation tends to zero and the sequence of approximations $x_0, x_1, x_2, \dots$ converges to the root ξ . Thus, when we express the equation $f(x) = 0$ in the form $x = \phi(x)$, then $\phi(x)$ must be such that

$$|\phi'(x)| < 1$$

in an immediate neighbourhood of the root. It follows that if the initial approximation x_0 is chosen in an interval containing the root ξ , then the sequence of approximations converges to the root ξ .

Now, we show that the root so obtained is unique. To prove this, let ξ_1 and ξ_2 be two roots of the equation $x = \phi(x)$. Then, we must have

$$\xi_1 = \phi(\xi_1) \quad \text{and} \quad \xi_2 = \phi(\xi_2)$$

Therefore,

$$|\xi_1 - \xi_2| = |\phi(\xi_1) - \phi(\xi_2)| = |\xi_1 - \xi_2| \phi'(\eta), \eta \in (\xi_1, \xi_2)$$

Hence

$$|\xi_1 - \xi_2| [1 - \phi'(\eta)] = 0$$

Since $|\phi'(\eta)| < 1$, it follows that $\xi_1 = \xi_2$, which proves that the root obtained is unique. Finally, we shall find the error in the root obtained. We have

$$|\xi - x_n| \leq k |\xi - x_{n-1}| = k |\xi - x_n + x_n - x_{n-1}| \leq k [| \xi - x_n | + |x_n - x_{n-1}|]$$

$$\Rightarrow |\xi - x_n| \leq \frac{k}{1-k} |x_n - x_{n-1}| = \frac{k}{1-k} k^{n-1} |x_1 - x_0| \leq \frac{k^n}{1-k} |x_1 - x_0|,$$

Which shows that the convergence would be faster for smaller values of k . Now, let ϵ be the specified accuracy so that

$$|\xi - x_n| \leq \epsilon$$

Then, it gives

$$|x_n - x_{n-1}| \leq \frac{1-k}{k} \epsilon,$$

which can be used to find the difference between two successive approximations (or iterations) to achieve a prescribed accuracy.

Example: Find a real root of the equation $x^3 = 1 - x^2$ on the interval $[0, 1]$ with an accuracy of 10^{-4} .

Solution: We rewrite the equation as

$$x = \frac{1}{\sqrt{x+1}}$$

Here

$$\phi(x) = \frac{1}{\sqrt{x+1}} = (x+1)^{-1/2}$$

Therefore,

$$\phi'(x) = -\frac{1}{2}(x+1)^{-3/2} = -\frac{1}{2\sqrt{(x+1)^3}} < 1 \text{ in } [0, 1]$$

Also,

$$\max |\phi'(x)| = \frac{1}{2\sqrt{8}} = \frac{1}{4\sqrt{2}} = k < 0.2$$

Therefore, it gives

$$|x_n - x_{n-1}| \leq \frac{1-0.2}{0.2} \epsilon = 4 \times 10^{-4} = 0.0004$$

Taking $x_0 = 0.75$, we find

$$x_1 = \frac{1}{\sqrt{1.75}} = 0.75593$$

$$x_2 = \frac{1}{\sqrt{1.75593}} = 0.75465$$

$$x_3 = \frac{1}{\sqrt{1.75465}} = 0.75493.$$

Now, $|x_3 - x_2| = 0.00028 < 0.0004$. Hence, the required root is 0.7549, correct to four decimal places.

Example: Find a real root, correct to three decimal places, of the equation $2x - 3 = \cos x$ lying in the interval $\left[\frac{3}{2}, \frac{\pi}{2}\right]$.

Solution: Rewrite the given equation as

$$x = \frac{1}{2}(\cos x + 3)$$

Here

$$|\phi'(x)| = \frac{1}{2}|\sin x| < 1, \text{ in } \left[\frac{3}{2}, \frac{\pi}{2}\right].$$

Choosing $x_0 = 1.5$, we obtain successively:

$$\begin{aligned} x_1 &= \frac{1}{2}(\cos 1.5 + 3) = 1.5354 \\ x_2 &= \frac{1}{2}(\cos 1.5354 + 3) = 1.5177 \\ x_3 &= \frac{1}{2}(\cos 1.5177 + 3) = 1.5265 \\ x_4 &= \frac{1}{2}(\cos 1.5265 + 3) = 1.5221 \\ x_5 &= \frac{1}{2}(\cos 1.5221 + 3) = 1.5243 \\ x_6 &= \frac{1}{2}(\cos 1.5243 + 3) = 1.5232 \\ x_7 &= \frac{1}{2}(\cos 1.5232 + 3) = 1.5238 \end{aligned}$$

Now, $|x_7 - x_6| = 0.0006 < 0.001$. Hence, the root, correct to three decimal places is 1.524.